

Comparative Analysis of Machine Learning Models for RDF-Based Mutagenicity Prediction

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Abstract

This project presents a machine learning framework for predicting the mutagenicity of chemical compounds using features extracted from an RDF knowledge graph. Molecule-level features extracted from the RDF representation are merged with labeled training and testing datasets. Three supervised machine learning algorithms, Support Vector Machine (SVM), Decision Tree (DT), and Linear Regression, are trained and evaluated. Their performances are compared using Accuracy, Precision, Recall, F1-score, classification reports, and confusion matrices.

1 Introduction

Mutagenicity prediction is an important task in cheminformatics for identifying chemical compounds capable of causing genetic mutations. Instead of relying solely on molecular descriptors, this work utilizes semantic information represented in RDF format to generate interpretable molecular features. The extracted features are used to train machine learning models for binary classification of mutagenic and non-mutagenic compounds.

1.1 Objectives

The primary objectives of this work are:

- Extract molecule-level features from RDF data.
- Merge the extracted features with the provided training and testing datasets.
- Train Linear Regression, Support Vector Machine, and Decision Tree models.
- Evaluate and compare model performance using multiple evaluation metrics.
- Save the generated feature matrix and model performance comparison results.

2 Dataset

The project utilizes three datasets:

- `mutag_compound_features.csv`

- Contains RDF-derived molecular features.
- **trainingSet.tsv**
 - Contains molecule identifiers and training labels.
- **testSet.tsv**
 - Contains molecule identifiers and testing labels.

The target variable is:

`label_mutagenic`

where

- 1 = Mutagenic
- 0 = Non-mutagenic

3 Methodology

3.1 Step 1: Loading Data

The training dataset, testing dataset, and RDF-derived molecular feature matrix are loaded into Pandas DataFrames for subsequent preprocessing and model development.

3.2 Step 2: Data Integration

The molecule identifier is standardized by renaming the `molecule` column to `bond`. The extracted RDF feature matrix is merged with both the training and testing datasets using a left join. Missing values produced during the merge operation are replaced with zero.

3.3 Step 3: Feature Engineering

The following preprocessing operations are performed:

- Separation of predictor variables and target labels.
- Removal of non-feature attributes:
 - `id`
 - `bond`
 - `label_mutagenic`
- One-hot encoding of categorical variables.
- Alignment of training and testing feature spaces to ensure identical feature dimensions.

3.4 Step 4: Linear Regression Training

A Linear Regression model is trained using the RDF-derived molecular feature matrix and the corresponding training labels. Although Linear Regression is fundamentally designed for predicting continuous numerical values, it is employed in this study as a baseline model for binary classification.

The trained model predicts continuous values for the testing dataset. These predictions are transformed into binary class labels using a threshold value of 0.5. Predictions greater than or equal to 0.5 are classified as mutagenic (1), whereas predictions below 0.5 are classified as non-mutagenic (0).

The resulting binary predictions are evaluated using Accuracy, Precision, Recall, F1-score, a classification report, and a confusion matrix.

3.5 Step 5: Support Vector Machine Training

A linear Support Vector Machine classifier is initialized using a linear kernel and a random seed of 42. The classifier is trained using the RDF-derived feature matrix and corresponding training labels. Predictions are generated for the testing dataset and evaluated using standard classification metrics.

3.6 Step 6: Decision Tree Training

A Decision Tree classifier is initialized with a maximum depth of five and a random seed of 42. The classifier is trained using the same feature matrix as the SVM model and evaluated using the testing dataset.

3.7 Step 7: Model Evaluation

The predictive performance of all three machine learning models is evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

The computed metrics are stored in a comparison table to facilitate quantitative analysis.

3.8 Step 8: Classification Report

Detailed classification reports are generated for Linear Regression, Support Vector Machine, and Decision Tree. Each report includes class-wise Precision, Recall, F1-score, and Support.

3.9 Step 9: Confusion Matrix

Confusion matrices are generated for all three models to visualize the number of:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

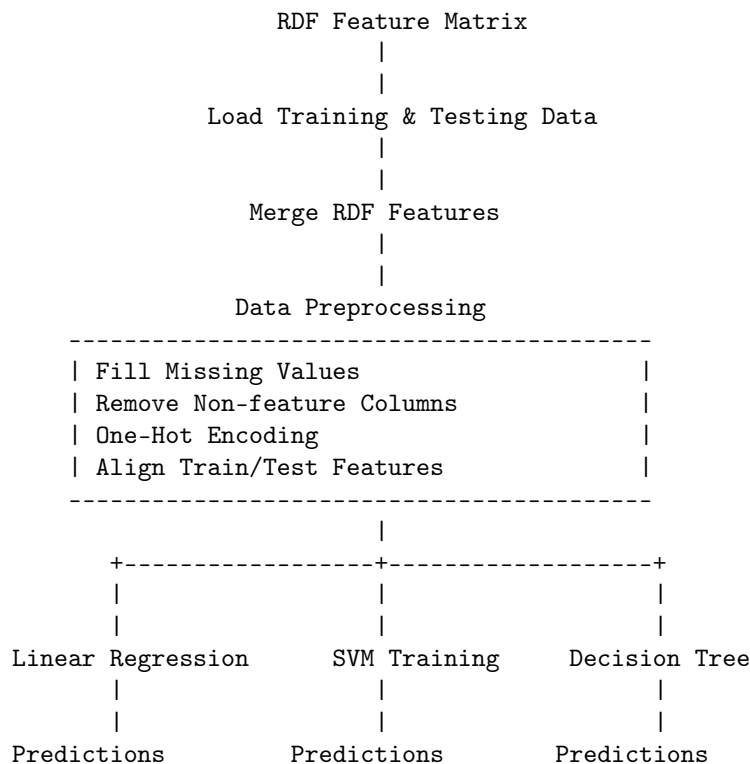
These matrices provide insight into the classification performance of each model.

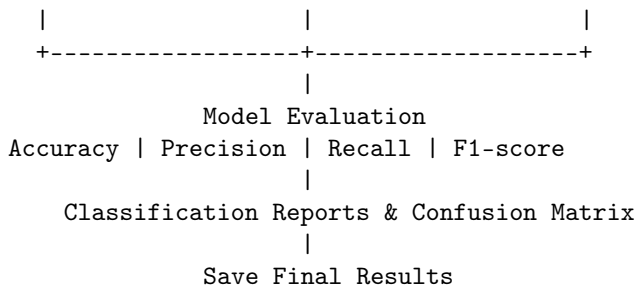
3.10 Step 10: Saving Results

The generated feature matrix and model performance comparison table are exported as CSV files:

- mutag_compound_features.csv
- model_performance_comparison.csv

4 Workflow





5 Machine Learning Algorithms

5.1 Linear Regression

Linear Regression is a supervised learning algorithm that models the relationship between predictor variables and a dependent variable using a linear equation. Although it is primarily intended for regression problems involving continuous outputs, it can also serve as a baseline approach for binary classification by applying a decision threshold to the predicted values.

In this work, Linear Regression predicts continuous scores for each chemical compound. These scores are converted into binary class labels using a threshold value of 0.5, enabling classification of compounds as mutagenic or non-mutagenic.

Advantages

- Simple and computationally efficient.
- Easy to implement and interpret.
- Serves as an effective baseline model.

Limitation

- Not specifically designed for classification tasks.
- May predict values outside the interval $[0,1]$.
- Generally provides lower classification performance than dedicated classifiers.

5.2 Support Vector Machine

Support Vector Machine constructs an optimal separating hyperplane that maximizes the margin between two classes. A linear kernel is employed because the RDF-derived molecular features form a high-dimensional sparse feature space.

Advantages

- Effective in high-dimensional feature spaces.
- Robust for binary classification.
- Strong generalization capability.

5.3 Decision Tree

Decision Trees classify compounds by recursively partitioning the feature space based on feature values. **Advantages**

- Easy to interpret.
- Fast training.
- Handles mixed feature types.

6 Evaluation Metrics

The following evaluation metrics are employed.

6.1 Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

6.2 Precision

$$Precision = \frac{TP}{TP + FP}$$

6.3 Recall

$$Recall = \frac{TP}{TP + FN}$$

6.4 F1-score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

7 Results

The performances of Linear Regression, Support Vector Machine, and Decision Tree are compared using:

- Accuracy
- Precision
- Recall
- F1-score

In the conducted experiments, all three models achieved identical evaluation metrics. This indicates that the RDF-derived feature representation provided a highly informative and well-separated feature space, enabling each algorithm to produce the same classification outcomes on the testing dataset.

Since Accuracy, Precision, Recall, and F1-score are computed directly from the predicted class labels, identical predictions naturally resulted in identical evaluation metrics. This observation suggests that the extracted RDF semantic features effectively captured the discriminative characteristics of the molecular compounds.

Classification reports and confusion matrices were further analyzed to verify that all three models produced equivalent prediction patterns.

8 Conclusion

This project demonstrates an end-to-end machine learning pipeline for predicting mutagenicity using RDF-derived molecular features. The proposed workflow includes RDF feature extraction, data integration, preprocessing, feature engineering, machine learning model training, performance evaluation, and result visualization.

Three machine learning algorithms—Linear Regression, Support Vector Machine, and Decision Tree—were implemented and compared using multiple evaluation metrics. Linear Regression served as a baseline model, while Support Vector Machine and Decision Tree provided classification-based approaches.

Experimental results indicated that all three models achieved identical predictive performance on the RDF-derived feature representation, suggesting that the extracted semantic features provided sufficient discriminatory information for the mutagenicity prediction task.

For future work, additional algorithms such as Logistic Regression, Random Forest, Gradient Boosting, XGBoost, or Graph Neural Networks. These approaches are more likely to reveal meaningful differences in predictive performance.

Contributions of Team Members

This project was completed as a solo effort. All aspects—including data analysis, feature extraction, model development, evaluation, explanation, and report preparation—were carried out independently by the author.

Acknowledgement

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